Joule Development Guide

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Joule | DEV

Confidential

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Messages

Messages can be used in dialog functions to display information to the user. In order to display dynamic information, SpEL as well as handlebars scripting can be used to generate the message. For more details on supported scripting, see Scripting.

Example in dialog function:

Sample Code

```
- actions:
    - type: message
      message:
        type: quickReplies
        content:
          title: For which city would you like to see the weather?
          buttons:
            - title: Walldorf
              value: Walldorf
            - title: Paris
              value: Paris
            - title: Shanghai
              value: Shanghai
            - title: Palo Alto
              value: Palo Alto
```

Text

1. Type the text message that should be your assistant's response to the user.
2. You can enable Markdown syntax to format your text. For more information, see https://commonmark.org/help/ .
3. You can add a delay of up to 5 seconds between each message in a group of messages.

Sample Code

```
- actions:
    - type: message
      message:
        type: text
        content: Hello User!
        markdown: false
        delay: 3
```

Cards

1. Provide the title, subtitle, and add an image URL (title and subtitle are mandatory).
2. Add an interactive button to:
    a. web_URL: Open a link.

> **Note**
>
> This button type has been deprecated starting 1.3.0 specification. Use the `navigation` button type instead.

b. Postback: Add a postback to send back to the /dialog API or a URI that is to be opened.

c. Phone Number: Provide a phone number. Tapping or clicking this number triggers a phone call using the default calling application. This feature is available for specific channels.

d. Navigation: Configure an external navigation. This button type supports intent-based navigation or can be used to open full URLs in a new tab.

3. You can add a delay of up to 5 seconds between each message in a group of messages.

You can also add additional details on the card like description, status text, and detail section.

Sample Code

```
- actions:
  - message:
      type: card
      delay: 3
      content:
        title: <title>
        subtitle: <subtitle>
        imageUrl: <image url>
        imageStyle: <default|avatar>
        enableDetailView: <true|false>
        buttons:
          - type: web_url
            title: <button title>
            value: <link url>
          - type: postback
            title: <button title>
            text: >-
              <optional text that should be displayed in the chat on user
              action>
            value: <postback value sent to the assistant for processing>
          - type: phonenumber
            title: <button title>
            value: <phone number>
```

You can also add additional details on the card like description, status text, and detail section.

Sample Code

```
actions:
  - message:
      type: card
      content:
        title: This is the card title
        subtitle: This is the subtitle
        imageUrl: 'sap-icon://product'
        imageStyle: avatar
        description:
          value: >-
            You can now add a **description** to the card for more information
            if needed
```

```yaml
            markdown: true
            linesInDetailView: 7
          status: Product
          statusState: success
          hideHeaderInDetailView: false
          sections:
            - title: Section 1 Title
              attributes:
                - label: Label (Text value)
                  value: >-
                    Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed
                    do eiusmod tempor incididunt ut labore et dolore magna aliqua.
                    Ut enim ad minim veniam, quis nostrud exercitation ullamco
                    laboris nisi ut aliquip ex ea commodo consequat. Duis aute
                    irure dolor in reprehenderit in voluptate velit esse cillum
                    dolore eu fugiat nulla pariatur. Excepteur sint occaecat
                    cupidatat non proident, sunt in culpa qui officia deserunt
                    mollit anim id est laborum.  Lorem ipsum dolor sit amet,
                    consectetur adipiscing elit, sed do eiusmod tempor incididunt
                    ut labore et dolore magna aliqua. Ut enim ad minim veniam,
                    quis nostrud exercitation ullamco laboris nisi ut aliquip ex
                    ea commodo consequat. Duis aute irure dolor in reprehenderit
                    in voluptate velit esse cillum dolore eu fugiat nulla
                    pariatur. Excepteur sint occaecat cupidatat non proident, sunt
                    in culpa qui officia deserunt mollit anim id est laborum.
                - label: Label (Link value)
                  value: SAP
                  url: 'https://www.sap.com'
                  type: link
                - label: Markdown Attribute
                  value: >-
                    **Answer 1:** When transferring within the group, for example
                    from SAP Germany SE to SAP SE, you need to know the following:
                    You will receive a new employment contract from the hiring
                    company.  You don't have to submit a workflow through
                    SuccessMap in case of a National Transfer. You have to submit
                    a signed (wet signature) resignation letter in original via
                    inhouse-mail or regular mail to the HR department (SAP SE,
                    PERS02, Dietmar-Hopp-Allee 16, 69190 Walldorf). Please note
                    that due to legal regulation for Germany a signed resignation
                    letter in original has to be sent to the employer. The
                    resignation letter should include a comment that it is a
                    national transfer.
                  markdown: true
          buttons:
            - type: postback
              title: Post back button
              text: Post back display text
              value: hi
              imageUrl: 'sap-icon://travel-request'
            - type: web_url
              title: Link button
              value: 'https://help.sap.com'
              imageUrl: 'sap-icon://product'
            - type: navigation
              title: Display Service Contracts
              navigation_target:
                ibnParams:
                  P_DisplayCurrency: EUR
                ibnTarget:
                  semanticObject: ServiceContract
                  action: displayFactSheet
```


Buttons

1. Type your assistant's message asking the user to choose an appropriate option.

2. Add an interactive button to as described in the Card section.

3. You can add a delay of up to 5 seconds between each message in a group of messages.

Sample Code

```
- actions:
    - type: message
      message:
      - type: buttons
        content: null
        title: <title>
        buttons:
          - type: <postback|web_url|clientdata|phonenumber|navigation> # web_url is deprecated
            title: <reply title>
            text: <display text> #used only when type is postback
            value: <reply value>
            client_data: null #defined only when type is clientdata
            <key>: <value>
            delay: 3
```

Example with additional button types:

Sample Code

```
- actions:
    - type: message
      message:
        - type: buttons
          delay: 0
          content:
            title: Button List
            buttons:
              - type: web_url
                title: Link
                value: 'https://help.sap.com'
                imageUrl: >-
                  https://logos-world.net/wp-content/uploads/2022/02/SAP-Logo.jpg
              - type: clientdata
                title: Button (Client Side Only)
                value: 'https://sap.com'
                client_data:
                  key1: Data for key1
                  key2: data for key2
              - type: postback
                title: "Post back"
                text: "Post back display text"
                value: "Post back value"
                imageUrl: 'sap-icon://learning-assistant'
              - type: phonenumber
                title: Call someone
                value: '1231231234'
              - type: navigation
                title: Open example.com
                navigation_target: null
                url: 'https://example.com'
              - type: navigation
                title: Display Service Contracts (IBN with fallback URL)
                navigation_target:
                  ibnParams:
                    P_DisplayCurrency: EUR
                  ibnTarget:
                    semanticObject: ServiceContract
                    action: displayFactSheet
                  url: <optional fallback URL>
```

Quick Replies

Provide the following information:

1. The assistant's message asking the user to choose an appropriate option (optional).

2. Quick reply text that will be displayed to the user. You can add up to 11 quick replies.

3. Quick reply value that is sent back for further action.

ℹ Note

Quick Replies should always be the last message in a response, else the options will not be displayed.

Sample Code

```
actionGroups:
  - actions:
    - type: message
      message:
        type: quickReplies
        content:
          title: <title>
          buttons:
            - title: <reply title>
              value: <reply value>
          delay: 3
```

Carousel

This is a list of basic cards aligned vertically. Follow the steps mentioned for creating a card.

Sample Code

```
- actions:
    - type: message
      message:
        type: carousel
        content:
          enableDetailView: <true|false>
          cards:
            - title: <title>
              subtitle: <subtitle>
              imageUrl: <image url>
              buttons:
                - type: postback
                  title: <button title>
                  text: <optional display text>
                  value: <reply value>
            - title: <title>
              subtitle: <subtitle>
              imageUrl: <image url>
              buttons:
                - type: phonenumber
                  title: <title>
                  value: <phone number>
        delay: 3
```

You can also add additional details on each card like description, status text, and detail section. Supports all button types defined for a button response type. You can use the same format that is used for creating a button.

Sample Code

```
- actions:
  - message:
      type: carousel
      delay: 0
      content:
```

```
cards:
  - title: Title 1
    subtitle: Here is the subtitle
    description: This is a description line here.
    buttons:
      - type: web_url
        title: Button
        value: 'https://sap.com'
      - type: postback
        title: Post back title
        text: Post back display text
        value: post back value
      - type: clientdata
        title: Button (Client Side Only)
        value: 'https://sap.com'
        client_data:
          key1: Data for key1
          key2: data for key2
    status: Status
    statusState: success
    imageUrl: >-
      https://www.sap.com/dam/application/shared/logos/sap-logo-svg.svg.adapt.svg/149303
    sections:
      - title: Section Title 1
        attributes:
          - url: 'https://www.sap.com'
            value: text
            type: link
            label: Item Label Link (Client_data)
          - type: clientdata
            label: Item Label ClientSideOnly (Client_data)
            value: 'https://sap.com'
            client_data:
              key1: Data for kay1
              key2: data for key2
  - title: Title 2
    subtitle: Subtitle
    description: >-
      You can now add a description to the card for more information if
      needed.
    imageUrl: >-
      https://www.sap.com/dam/application/shared/logos/sap-logo-svg.svg.adapt.svg/149303
    hideHeaderInDetailView: true
    buttons:
      - value: demo 1
        title: Postback
        type: postback
```

List

1. Add a title, subtitle, and an image URL (all optional).
2. For each list item, add a title (mandatory), subtitle, description, and an image URL. Only the title is mandatory.

3. Add a button on each list item for the users to make a selection. Supports all button types defined for a button response type. You can use the same format that is used for creating a button.

4. You may add a button for the overall list, for example, to open the full list in your application or website.

5. You can choose to display the list in a collapsible panel container.

6. You can add a delay of up to 5 seconds between each message in a group of messages.

Sample Code

```
- actions:
    - type: message
      message:
        type: list
        content:
          title: <title>
          subtitle: <subtitle>
          enableDetailView: <true|false>
          imageUrl: <image url>
          imageStyle: <default|avatar>
          elements:
            - title: <element title>
              subtitle: <element subtitle>
              imageUrl: <element image url>
              imageStyle: <default|avatar>
              description: <element description text>
              buttons:
                - type: postback
                  title: <reply title>
                  text: <optional display text>
                  value: <reply value>
        delay: 3
```

Example with a detail section:

Sample Code

```
- actions:
  - message:
      type: list
      content:
        title: Header Title
        total: 13
        subtitle: Subtitle
        description: This is the description line.
        buttons:
          - type: postback
            title: Post back title
            value: Post back value
        elements:
          - title: Title 1
            subtitle: Subtitle 1
            imageUrl: >-
              https://www.sap.com/dam/application/shared/logos/sap-logo-svg.svg.adapt.svg/149303
            imageStyle: avatar
            buttons:
              - type: web_url
                title: Button 1 (link)
                value: 'https://google.com'
```

```
          - type: postback
            title: Button 2 (Postback)
            text: Postback display text
            value: hi
          - type: clientdata
            title: Button (Client Side Only)
            client_data:
              ItemClientOnly1: Data for item one
              ItemClientOnly2: data for item 2
          - type: phonenumber
            title: Button 3 (Phone)
            value: '11231231234'
          - type: navigation
            title: Display Service Contracts (IBN without fallback URL)
            navigation_target:
              ibnParams:
                P_DisplayCurrency: EUR
              ibnTarget:
                semanticObject: ServiceContract
                action: displayFactSheet
      - title: Title 2
        subtitle: Subtitle 2
        description: This is a description line (3 buttons)
        buttons:
          - type: web_url
            title: Button 1 (link)
            value: 'https://google.com'
      - title: Title 3
        subtitle: Subtitle 3
        buttons:
          - type: postback
            title: Post back title
            value: Post back value
        sections:
          - title: Section 1 Title
            attributes:
              - label: Label (Text value)
                value: Display text
              - url: 'https://www.sap.com'
                type: link
                value: text
                label: Label (Link value)
              - label: Label 3
                value: This is text 3
      - title: Title 4
        subtitle: Subtitle 4
        description: Here is the description.
```

Example for a list displayed in a panel.

Sample Code

```yaml
- actions:
    - type: message
      message:
        - type: list
          delay: 0
          content:
            displayInPanel: true
            panel:
              collapsed: true
            title: Search results
            enableDetailView: false
            elements:
              - imageUrl: 'sap-icon://question-mark'
                title:
                  value: >-
                    [Create Sales Orders Using Automatic
                    Extraction](https://google.com)
                  markdown: true
                description: >-
                  Create Sales Orders - Automatic extraction App 10. +4920 with
                  this app, you as an internal sales representative can create
                  sales orders from purchase order files in PDF or image which
                  is a built-end feature.
              - title:
                  value: >-
                    [Create Sales Orders Automatic
                    Extraction](https://google.com)
                  markdown: true
                imageUrl: 'sap-icon://activatesap-icon://learning-assistant'
                description: >-
                  Create Sales Orders - Automatic extraction App 10. +4920 with
                  this app, you as an internal sales representative can create
                  sales orders from purchase order files in PDF or image which
                  is a built-end feature.
              - title:
                  value: >-
                    [Create Sales Orders with PDF Automatic
                    Extraction](https://google.com)
                  markdown: true
                imageUrl: 'sap-icon://customer'
                description: >-
                  Create Sales Orders - Automatic extraction App 10. +4920 with
                  this app, you as an internal sales representative can create
                  sales orders from purchase order files in PDF or image which
                  is a built-end feature.
```

Media

Provide a URL of an image or video that should be displayed to the user. You can also add a delay of up to 5 seconds between each message in a group of messages.

Limitations:

- URLs for image or video should be public.
- Supported URLs:
  - The supported video formats are - *MP4*, *WEBM*, *OGV*, and *3GP*.
  - The video URLs that are currently supported are - *YouTube*, *Vimeo*, *Daily Motion*, *SAP Video*, and *SAP TV Video*.

Sample Code

```
- actions:
    - type: message
      message:
        type: media
        content: <media url>
        delay: 3
```

Client Data

The content of client data messages is not displayed directly on the screen. This message type allows you to define a JSON key-value pair that is returned to the web client.

Sample Code

```
- actions:
    - type: message
      message:
        type: client_data
        content:
          elements:
            client_action:
              fiori:
                navigate:
                  semanticObject: <semanticObject>
                  action: <action>
                  params:
                    key1: value1
                  appStateData: {}
```

UI Integration Card

Add a UI integration card `manifest` to the content to display a non-actionable UI5 integration card.

Sample Code

```
- actions:
    - type: message
      message:
        type: ui5integrationCard
        content:
          manifest: <UI Integration card manifest>
          submit_message: >-
            <Optional submit message to be sent in the chat on Submit of an
            interactive integration card>
        delay: 3
```

Illustrated Message

Provide a URL of an image or video that should be displayed to the user. You can also add **Define an illustrated message** in scenarios where you would like to show a success message or an error message as a result of a transaction. You can also use this message to display a no-data state to your users with a nuanced actions to guide the user to the next step in the conversation. Navigation can be added via call to action buttons to help user keep track of important actions they can take outside of the assistant or within the surrounding application where your assistant is integrated.

Illustrated messages are informative cards consisting of illustrations, a header title, a subtitle and up to two call to action buttons.

Sample Code

```
- actions:
    - type: message
      message:
        - type: illustrated_message
          content:
            illustration_name: tntSuccess
            title: This is the illustrated_message title
            subtitle: This is the illustrated_message subtitle
            buttons:
              - title: <button title>
                type: clientdata
                client_data:
                  <key>: <value>
              - title: <button title>
                type: postback
                text: <display text>
                value: <postback value>
```

Likert Scale

Define the Likert Scale message for a Likert Slot Filling option in the scenario:

Sample Code

scenarios/joule_satisfaction.yaml

```
description: This scenario captures the user's satisfaction level with the Joule platform.
slots:
  - name: joule_fit_for_purpose
    description: The user's perception of whether the Joule platform does what they need it to d
target:
  type: function
  name: scenario_joule_survey
```

The likert_scale message type enables Joule Content Developers to configure pointer scales, labels, and values at design time. It aims to ensure proper rendering of Likert scales in the Joule Web Client.

The supported range is from 1 to 7. The compiler will not enforce validation for contiguous values (i.e., allowing gaps between Likert option values) or prevent value duplication (i.e., permitting multiple options with the same value). This approach provides survey designers with greater flexibility to configure their Likert scales according to specific requirements.

Sample Code

functions/scenario_joule_survey.yaml

```
type: message
message:
    type: likert_scale
    content: # (Required) Main configuration block for the Likert scale element
        prompt: 'Joule does what I need it to do'
        options: # Required; array of scale options. Minimum: 1, Maximum: 7 items.
            - label: 'Strongly disagree' # (Required) Text shown on the button for this option.
              value: 1
            - label: Disagree
              value: 2
            - label: Neutral
              value: 3
            - label: Agree
              value: 4
            - label: 'Strongly agree'
              value: 5
```

If the user select 4, then the slot *joule_fit_for_purpose* should now contain the value 4 and be usable in downstream parts of the dialog function, i.e:

Sample Code

Usage Example of the Likert Slot Filling Parameter

```
- condition: joule_fit_for_purpose != null
  actions:
    - type: api-request
      method: POST
      system_alias: JouleFeedbackService
      path: api/joule-feedback
      headers:
        Content-Type: application/json
      body: { "fit_for_purpose": "{{joule_fit_for_purpose}}" }
    - type: message
      scripting_type: handlebars
      message:
        type: text
        content: "Thank you for your feedback on Joule! You rated its fit for purpose as {{joule
```

UI Integration Cards

You can define a UI integration card manifest as a response in your assistant.

A UI integration card can be used to respond to the user with a non-actionable card that presents data in a rich UI format, using different card types.

For more information about the available card types, see UI Integration Card.

Data Binding

You can use scripting to bind context data in your assistant to the UI Integration card. For more information, see Scripting. You can define the data binding in the following ways:

- Bind the context data to the `json` property of UI5 integration card and use the UI Integration card binding syntax for the individual UI elements defined in the manifest.
- Use the DA binding syntax to bind context directly to individual UI elements defined in the manifest.

ℹ Note

Dynamic data binding to the card using destination configurations inside the manifest is not supported. Use Dialog Functions to make remote API calls and bind the data as defined above.

- ### Sample Code

```
type: ui5integrationCard
content:
  submit_message: >-
    Request product update of <? $mock_data.product.ProductName ?> to
    {submitted_data.ProductName}
  manifest: |
    {
                "_version": "1.17.0",
                "sap.app": {
                  "type": "card",
                  "id": "card.explorer.adaptive.extensionSample"
                },
                "sap.card": {
                  "data": {
                    "json": "<? $product_data ?>"
                  },
                  "type": "AdaptiveCard",
                  "header": {
                    "title": "Edit Product"
                  },
                  "content": {
                    "schema": "http://adaptivecards.io/schemas/adaptive-card.json",
                    "type": "AdaptiveCard",
                    "version": "1.0",
                    "body": [
                      {
                        "type": "Input.Text",
                        "label": "Product Name",
                        "value": "${product.ProductName}",
                        "style": "text",
                        "id": "ProductName"
                      },
                      {
                        "type": "Input.ChoiceSet",
                        "label": "Product Category",
                        "id": "CategoryID",
                        "style": "compact",
                        "isMultiSelect": false,
                        "value": "${product.CategoryID}",
```

```
                "choices": [
                  {
                    "data": "${categories}",
                    "title": "${CategoryName}",
                    "value": "${CategoryID}"
                  }
                ]
              },
              {
                "type": "Input.Text",
                "label": "Supplier Name",
                "value": "<? $mock_data.product.supplier.CompanyName ?>",
                "style": "text",
                "id": "CompanyName"
              }
            ],
            "actions": [
              {
                "type": "Action.Submit",
                "title": "Submit"
              }
            ]
          }
        }
```

Card Actions

For information about card action types, see Card Actions.

ℹ Note

- The **navigation action type** (`Navigation`) supports only absolute URLs that open in a new tab.
- The custom action type (`Custom`) is not supported.

Restrictions

- If you are using the `data` property of the integration card to bind data to the card, only the static data binding json is supported.
- Action Handlers inside the manifest are not supported.
- Component card types cannot be defined as a response in your assistant.
- Filters and user defined manifest parameters are not supported.
- Dynamic data fetching on the front end via destination declarations in the manifest are not supported.

Configure Cross-Product Navigation

You can configure navigation to business applications or external application in Joule responses. This helps in designing conversational scenarios that allows end users to navigate to specific applications that provide additional context to the ongoing conversation or complete complex transactions inside the application that is launched.

A button type called `navigation` is supported to configure such interactions. This button type is supported within the following UI Element responses within Joule.

- Buttons
- Cards
- List
- Carousel
- UI Integration Cards support Navigation action out of the box. See format.

Setup for Cross Product Navigation

For information on the prerequisites for setting up cross product navigation, see Cross-Product Navigation. Joule platform uses the Navigation service to resolve the IBN targets defined in capabilities.

ℹ Note

Earlier, Joule required the capability developers to implement logic to call navigation service via dialog functions, resolve the required navigation target to a valid `url` and return it along with the ibn target and parameters using the `url` property defined for navigation targets. This is no longer required and as a mandatory guideline it is requested to remove the `url` property from the capabilities. This property is removed starting DT Spec version 3.20. With the removal of the `url` property, resolution of navigation targets via direct calls to the Navigation Service within Dialog Functions in capabilities is no longer allowed.

Until the removal of the property, the `url` continues to be maintained in the capability, Joule platform will not resolve the target, instead navigate to the provided `url` on web. For Joule on mobile, the targets will be resolved for the mobile device type.

Definition of Navigation Button Type

Attribute	Type	Description	Required	Allowed Values	Limitations	Since Version
type	string	type of the button	true	navigation		1.4.0
navigation_target	navigation_target (see Navigation Target).	Represents the navigation target that should be launched when the button is clicked. This case can be used to perform either an intent based navigation or navigation to a URL in a new tab.	true			1.4.0

Navigation Target

Use this to define a configuration for the navigation action and perform either an intent based navigation or navigation to a URL in a new tab.

- **Define an intent based navigation**

  The intent-based navigation mechanism allows launching applications in-place using semantic objects and actions. For this to work, you must maintain the exact parameters, and ensure that the host application where Joule is embedded supports in-place navigation to intents.

  Note

  In-place intent based navigation is supported by Joule in SAP S/4HANA (Fiori Launchpad). For other business applications, check the documentation for onNavigation callback in Web Client bridge. If the callback is not implemented by the host application, Joule will always open the resolved url in a new tab.

Attribute	Type	Description	Required	Allowed Values	Limitations	Since Version
ibnTarget	object (see Properties for intent based navigation targets)	Intent-based navigation target	true	navigation		1.4.0
ibnParams	object	Map of parameters for intent-based navigation.	false			1.4.0
url **Note** This url will be deprecated from version 3.20.	string	Optional fallback URL which Joule uses when the intent cannot be resolved by the framework. **Note** Earlier, Joule required the capability developers to implement logic to call navigation service via dialog functions, resolve the required navigation target to a valid URL and return it along with	false	https URL or relative URL		1.4.0

Attribute	Type	Description	Required	Allowed Values	Limitations	Since Version
		the ibn target and parameters using the `url` property defined for navigation targets. This is no longer required and as a mandatory guideline it is requested to remove the `url` property from the capabilities. This property will be deprecated and removed in Q2 2025. Until the removal of the property is the `url` continues to be maintained in the capability, Joule platform will not resolve the target, instead navigate the the provided `url` on web. For Joule on mobile, the targets will be resolved for the mobile device type.				

Properties for intent based navigation targets

Attribute	Type	Description	Required	Since Version
semanticObject	string	Semantic object for the navigation target.	true	1.4.0
action	string	Action configuration of the navigation target.	true	1.4.0

- **Define URL navigation**

Attribute	Type	Description	Required	Allowed Values	Since Version
url	string	URL must be launched in a new tab.	true	https URL or relative URL.	1.4.0

Additional Information

Custom handling of navigation

If you want to integrate Joule with an SAP business application, you have the option to handle navigation yourself using Web Client bridge APIs.

You can implement platform dependent navigation logic, either using the URL (if resolved) or the IBN targets in the `onNavigation` (payload) API of your Web Client bridge. If such an implementation is provided in your bridge, Web Client will call this method on user action, instead of executing the default logic. For more information, see Web Client bridge API documentation .

Examples

Intent based navigation

Sample Code

```
- type: navigation
  title: Display Service Contracts
  navigation_target:
    ibnParams:
      P_DisplayCurrency : EUR
    ibnTarget:
      semanticObject: ServiceContract
      action: displayFactSheet
```

Intent-based navigation for UI Integration Cards

To use intent-based navigation (IBN) in UI integration cards, capability developers must ensure that only Joule data binding syntax is used in the manifest definition. This ensures that the manifest is fully resolved by Joule platform to determine IBN target definitions and their values.

Note

UI5 binding syntax along with IBN is not supported in UI integration cards.

URL navigation

Sample Code

```
- type: navigation
  title: Go to Example.com
  navigation_target:
    url: https://example.comx
```

Guidelines

For additional details on navigation behavior in Joule and guidelines to be followed, see Guidelines for Capability Development.

UI Constraints and Limits

The following table lists the various components of UI Constraints and Limits and details of each component.

Text Length Limits

Component	Attribute	Limit in Web Client (Default width)	Limit in Web Client (Full Screen)	Comments
Text	content (markdown: false)	MAX_SAFE_INTEGER		Limit to the javascript MAX_SAFE_INTEGER value
	content (markdown: true)	MAX_SAFE_INTEGER		
Card	title (markdown: false)	Displays up to 3 lines of available space. Maximum 640 characters		
	title (markdown: true)	Approximately 150 characters	Approximately 240 characters	Excluding the markdown symbols
	subtitle (markdown: false)	Displays up to 2 lines of available space. Maximum 640 characters		
	subtitle (markdown: true)	Approximately 100 characters	Approximately 160 characters	Excluding the markdown symbols
	description (markdown: false)	Displays up to 5 lines of available space. Maximum 640 characters		
	description (markdown: true)	Approximately 250 characters	Approximately 400 characters	Excluding the markdown symbols
Buttons (Message)	title	80 characters		

Component	Attribute	Limit in Web Client (Default width)	Limit in Web Client (Full Screen)	Comments
Button (individual button displayed in Buttons, List, Carousel and Card)	title	640 characters		
	text	80 characters		
Quick Reply	Message title (markdown: false)	640 characters		
	Message title (markdown: true)	Approximately 640 characters		
	text	32 characters		
List	Header title (markdown: false)	Displays up to 3 lines of available space. Maximum 640 characters		
	Header title (markdown: true)	• 150 characters without image and row count • 132 characters with image • 90 characters with row count • 72 characters with image and row count	• 240 characters without image and row count • 220 characters with image • 180 characters with row count • 160 characters with image and row count	
	Header subtitle (markdown: false)	Displays up to 2 lines of available space. Maximum 640 characters		
	Header subtitle (markdown: true)	• 100 characters without image • 88 characters with image	• 160 characters without image • 148 characters with image	
	Header description (markdown: false)	Displays up to 5 lines of available space. Maximum 640 characters		
	Header description (markdown: true)	• 250 characters without image • 220 characters with image	• 400 characters without image • 370 characters with image	
	List element title (markdown: false)	Displays up to 3 lines of available space.		

Component	Attribute	Limit in Web Client (Default width)	Limit in Web Client (Full Screen)	Comments
		Maximum 640 characters		
	List element title (markdown: true)	• 150 characters without image and status or button • 132 characters with image • 90 characters with status or button • 72 characters with image and status or button	• 240 characters without image and status or button • 220 characters with image • 180 characters with status or button • 160 characters with image and status or button	
	List element subtitle (markdown: false)	Displays up to 2 lines of available space. Maximum 640 characters		
	List element subtitle (markdown: true)	• 100 characters without image and button • 88 characters with image • 60 characters with button • 48 characters with image and button	• 160 characters without image and button • 148 characters with image • 120 characters with button • 108 characters with image and button	
	List element description (markdown: false)	Displays up to 5 lines of available space. Maximum 640 characters		
	List element description (markdown: true)	• 250 characters without image and button • 220 characters with image • 150 characters with button • 120 characters with image and button	• 400 characters without image and button • 370 characters with image • 300 characters with button • 270 characters with image and button	

Component	Attribute	Limit in Web Client (Default width)	Limit in Web Client (Full Screen)	Comments
	List element subtitle (markdown : true)	~ 100 without image and button ~ 88 with image ~60 with button ~ 48 with image and button	~ 160 without image and button ~ 148 with image ~120 with button ~ 108 with image and button	
Carousel	Carousel contains a set of cards displayed horizontally. See Card limits			
Detail View (displayed in Card, List and Carousel)	Detail View title (markdown: false)	Displays up to line(s) defined in `linesInDetailView` (Defaults 3) of available space. Maximum 640 characters		Values taken over from definition in the individual element of the composite control (that is. Card, List element, Carousel card). Can be controlled using the attribute `linesInDetailView`. If `linesInDetailView` is not defined, will use the default card limits For Carousel, the text lengths will ignores the image
	Detail View title (markdown: true)	~ 150 without image and status ~ 132 with image ~ 90 with status ~ 72 with image and status if `linesInDetailView` is defined, see the note below	~ 240 without image and status ~ 220 with image ~180 with status ~ 160 with image and status if `linesInDetailView` is defined, see the note below	
	Detail View subtitle (markdown: false)	Displays up to line(s) defined in `linesInDetailView` (Defaults 2) of available space. Maximum 640 characters		
	Detail View subtitle (markdown: true)	~ 100 without image ~ 88 with image if `linesInDetailView` is defined, see the note below	~ 160 without image ~ 148 with image if `linesInDetailView` is defined, see the note below	
	Detail View description (markdown: false)	Displays up to line(s) defined in `linesInDetailView` (Defaults 5) of available space. Maximum 640 characters		

Component	Attribute	Limit in Web Client (Default width)	Limit in Web Client (Full Screen)	Comments
	Detail View description (markdown: true)	~ 250 without image ~ 220 with image if `linesInDetailView` is defined, see the note below	~ 400 without image ~ 370 with image if `linesInDetailView` is defined, see the note below	
	Section header title	640 characters		
	Section attribute label	Displays up to 1 line of available space		
	Section attribute value (markdown: true)	Displays up to 4 lines of available space		
	Section attribute value (markdown: false)	Approximately 200 characters	Approximately 320 characters	

ℹ Note

If the `linesInDetailView` is defined for markdown text, the following logic is used:

maxLength = linesInDetailView * (defaultLength - imageLength - statusLength)

Standard: defaultLength = 50, imageLength = 6, statusLength = 20

Full Screen: defaultLength = 80, imageLength = 6, statusLength = 20

The imageLength and or statusLength will be 0 if no image or status is defined.

UI Elements Limit

Component	Elements	Minimum	Maximum	Comments
Card	button	1	3	
	section	0	1	
	content or attribute pairs	5		
List	button	0	1	
	elements	1	30	6 elements are shown initially. **View More** allows the user to see additional elements upto 30 elements. Following attributes can be used to control: • initial page size (initialPageSize) • incremental page size (incrementPageSize)

Component	Elements	Minimum	Maximum	Comments
				• Maximum elements in a list should be less than 30 (maxNumberOfRowsInList)
	element buttons	0	7	
Buttons	button	1	15	5 buttons are shown initially. **View More** allows the user to see additional buttons upto 15.
Quick Replies	button		11	
Carousel	card	1	12	After the pages reach the count of 8, the dots are replaced by numeral representation (valid for all devices).

Markdown Limitations

Joule supports extended markdown syntax in responses except for the following limitations:

- Heading IDs - Not Supported, for more information, see Heading IDs .
- Definition Lists - Not Supported, for more information, see Definition Lists .
- Emoji Shortcodes - Not Supported, for more information, see Using Emoji Shortcodes .
- Word Highlight - Not Supported, for more information, see Highlight .